IIT Roorkee

Technical Communication Lab

Question 1 (10 Marks)

Topic: Recursion and Function Call Stack

Instructions:

- Write your answer entirely in LaTeX.
- Use proper environments: `itemize`, `tabular`, and math mode.
- Answers without proper LaTeX formatting will lose marks.

Question:

Let

`arr = {1,2,3,4,5,6,7,8}`

```
void foo(int arr[], int a, int b, int len) {
    for (int i = 0; i < len; i++) {
        int temp = arr[a + i];
        arr[a + i] = arr[b + i];
        arr[b + i] = temp;
    }
}

void fun(int arr[], int l, int r) {
    if (l >= r) return;

    int mid = (l + r) / 2;

    fun(arr, l, mid);
    fun(arr, mid + 1, r);

    int left = mid - l + 1;
    int right = r - mid;
    int len = (left < right) ? left : right;

    foo(arr, l, mid + 1, len);
}
```

Evaluate:

`fun(arr, 0, 7)`

(a) **Recursion Tree (3 marks)**
Write the recursion tree using `itemize` environment in LaTeX.

(b) **Intermediate Arrays (4 marks)**
Write all intermediate arrays after each `foo()` call using:

$$\{\cdot, \cdot, \cdot\}$$

Each step must be written in a separate displayed equation.

(c) **Final Array (1 mark)**
Write the final array using proper math notation.

(d) **Stack Depth (1 mark)**
Write the maximum recursion depth in math mode.

(e) **Total Recursive Calls (1 mark)**
Write the total number of recursive calls.

Marks: (a) 3 (b) 4 (c) 1 (d) 1 (e) 1

Question 2

Write the LaTeX code for each of the following expressions.Properly structure your answers using mathematical environments.

a) Differentiation (2 Marks)

Write the LaTeX code for:

$$f(x) = \frac{x^3 - 3x + 2}{\sqrt{x^2 + 1}}$$

Also write the LaTeX code for its derivative.

Solution

$$f(x) = \frac{x^3 - 3x + 2}{\sqrt{x^2 + 1}}$$

Using quotient rule:

$$f'(x) = \frac{u'v - uv'}{v^2}$$

$$u' = 3x^2 - 3, \quad v' = \frac{x}{\sqrt{x^2 + 1}}$$

$$f'(x) = \frac{(3x^2 - 3)\sqrt{x^2 + 1} - (x^3 - 3x + 2)\frac{x}{\sqrt{x^2+1}}}{x^2 + 1}$$

$$\boxed{f'(x) = \frac{2x^4 + 3x^2 - 2x - 3}{(x^2 + 1)^{3/2}}}$$

b) Integration (2 Marks)

Write the LaTeX code for:

$$\int_1^e x \ln x \, dx$$

Also write the solution using integration by parts.

Solution

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$$

$$\int_1^e x \ln x \, dx = \left[\frac{x^2}{2} \ln x - \frac{x^2}{4} \right]_1^e$$

$$= \frac{e^2}{2} - \frac{e^2}{4} + \frac{1}{4}$$

$$\boxed{\int_1^e x \ln x \, dx = \frac{e^2+1}{4}}$$

c) Algebra (1 Marks)

Write the LaTeX code for:

$$2x^2 - 5x + 3 = 0$$

Also solve using factorisation and quadratic formula.

Solution

$$(2x-3)(x-1) = 0$$

$$x = \frac{3}{2}, \quad x = 1$$

Using quadratic formula:

$$x = \frac{5 \pm \sqrt{25-24}}{4}$$

$$\boxed{x = \frac{3}{2}, \quad x = 1}$$

Question 3 (5 Marks)

For each of the following sub-questions, write the complete LaTeX code for the given expressions and their step-by-step solutions.

(a) Determinant and Inverse of Matrix

Task: Write the LaTeX code for the matrix, determinant calculation, cofactors, and inverse.

Given:

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & 2 & 1 \end{pmatrix}$$

Solution

Step 1: Determinant

$$\det(A) = 2 \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} + 0$$

$$= 2(3 \cdot 1 - 2 \cdot 2) - 1(1 \cdot 1 - 2 \cdot 0) = 2(-1) - 1 = -3$$

$$\boxed{\det(A) = -3}$$

Step 2: Cofactors (example)

$$C_{11} = -1, \quad C_{12} = -1, \quad C_{13} = 2$$

Step 3: Inverse

$$A^{-1} = \frac{1}{-3} \operatorname{adj}(A)$$

$$\boxed{A^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 & -2 \\ 1 & -2 & 4 \\ -2 & 4 & -5 \end{pmatrix}}$$

(b) System of Equations (Cramer's Rule)

Task: Write the LaTeX code for the system and solve using Cramer's Rule.

$$\begin{aligned} 2x + y &= 5 \\ 3x - 2y &= 4 \end{aligned}$$

Solution

$$D = -7, \quad D_x = -14, \quad D_y = -7$$

$$x = \frac{D_x}{D} = 2, \quad y = \frac{D_y}{D} = 1$$

$$x = 2, \quad y = 1$$

(c) Vector Operations

Task: Write the LaTeX code for dot product, cross product, and angle.

$$\vec{u} = (1, 2, 3), \quad \vec{v} = (4, 5, 6)$$

Solution

$$\vec{u} \cdot \vec{v} = 32$$

$$\vec{u} \times \vec{v} = (-3, 6, -3)$$

$$\theta \approx 12.93^\circ$$

Final results: Dot = 32, Cross = (-3,6,-3), Angle ≈ 12.93°

(d) Eigenvalues and Eigenvectors

Task: Write the LaTeX code for eigenvalues and eigenvectors.

$$B = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

Solution

$$\lambda^2 - 7\lambda + 10 = 0 \Rightarrow \lambda = 5, 2$$

$$\vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \vec{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

Eigenvalues: $5, 2$
Eigenvectors: $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \end{pmatrix}$